Research Methodology in Computer Science

UNIT I WORKSHEET

Introduction to Research Problem



Course: CSEG3060 | Unit: I | Lecture Allocation: 9 Hours
Designed as an active-learning revision and practice worksheet.

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Section: B1[CCVT] **Date:** 15/9/2026

How to use this worksheet

This worksheet is meant to move you from *"I know the definitions"* to *"I can formulate and defend a research problem."* Work through the activities in order. Use your own Computer Science topic whenever a question asks you to design or formulate something.

Unit I Coverage Checklist

- ☑ Meaning and significance of research
- ☑ Objectives of research
- ☑ Key elements of research
- ☑ Sources of research problems
- ☑ Identification of research problems
- ☑ Good research problem criteria
- ☑ Common selection errors
- ☑ Avoiding selection errors
- ☑ Scope and boundaries
- ☑ Research objectives
- ☑ SMART objectives
- ☑ Research approaches
- ☑ Theoretical vs. empirical research
- ☑ Data types and sources
- ☑ Data collection tools
- ☑ Data collection guidelines
- ☑ Data analysis
- ☑ Interpretation of results
- ☑ Research instrumentation
- ☑ Calibration and repeatability
- ☑ Integrated case-based application

1 Part A — Research: What, Why and How?

A1. Research in one sentence

Research is a **systematic and scientific investigation** used to discover new facts, verify existing knowledge, analyze relationships, explain phenomena, and answer questions using evidence.
Key elements from class:

Problem Identification → Data Collection & Analysis → Hypothesis Formation → Result Interpretation

Activity 1: Build the research chain

Put the following in a sensible order:

Result Interpretation	Problem Identification	Data Collection/Analysis	Hypothesis Formation
4	1	2	3

Why is this order useful in a CS research project? **[2 marks]**

Each stage feeds the next: a clear problem guides data collection, analysis enables a testable hypothesis, and interpretation turns results into an answer to the problem.

Activity 2: Why does research matter?

Give one CS example for each:

Technology/knowledge advancement	Transformer models advancing natural-language processing.
Industrial innovation	Recommender systems boosting e-commerce sales.
Academic growth	New graph algorithms published and cited in later work.
Policy or societal decision-making	Predictive models guiding public-health resource planning.
Computer Science example (AI/algorithms/HCI/etc.)	ML-based intrusion detection with fewer false positives.

Activity 3: Research objectives

Complete the table.

Objective of research	Your CS example
Find answers using scientific methods	Measure how CNN depth affects image-classification accuracy.
Explain phenomena / establish relationships	Study how dataset size relates to model overfitting.
Test or validate an existing theory	Empirically test whether dropout reduces overfitting.

Think like a researcher

A student says: *"I want to build an AI app because AI is trending."* Is this already a research problem? Explain what is missing.

No. It states a trend-driven wish, not a problem. It lacks a specific gap, a research question, defined scope, measurable objectives, and success criteria - it says what to build, not what to investigate.

2 Part B — Types and Approaches of Research

Research approaches covered in Unit I

Exploratory
discover patterns/ideas

Descriptive
describe facts/characteristics

Analytical
analyze information

Theoretical
models, logic, proofs

Empirical
observation/experiment

The class material also introduced **fundamental** and **applied** research as broad research types. Choose the approach according to the research question and the kind of evidence required.

Activity 4: Match the approach

Write E (Exploratory), D (Descriptive), A (Analytical), T (Theoretical), or EM (Empirical).

1. Investigating an emerging problem where very little is known: E
2. Measuring average response time of five search algorithms: EM
3. Developing a formal model for a new computational system: T
4. Summarizing user characteristics from a usability survey: D
5. Comparing existing datasets and identifying relationships among variables: A

Activity 5: Theoretical or empirical?

For each scenario, decide the dominant approach and justify it in one sentence.

Scenario	Type	Why?
Prove a property of an algorithm mathematically	Theoretical	Uses formal logic and mathematical proof, not data.
Benchmark an intrusion-detection model on a dataset	Empirical	Conclusions drawn from measured experimental results.
Develop a formal model of a distributed protocol	Theoretical	Builds an abstract model rather than collecting data.
Interview developers to discover pain-points in an emerging tool	Empirical	Gathers first-hand qualitative data via interviews.

3 Part C — Finding and Formulating a Research Problem

C1. Where do research problems come from?

Important sources highlighted in class include:

- Literature reviews and research gaps.
- Industrial problems and real-world needs.
- Emerging trends and technologies.
- Academic coursework and past projects.
- Surveys of recent publications.
- Patent and research databases.
- Industry white papers.
- Competitive research proposals.

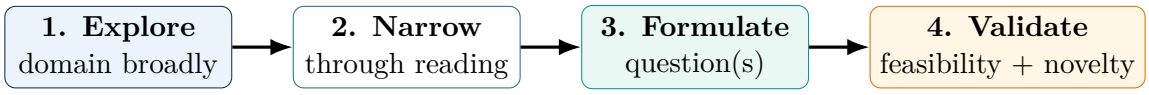
Activity 6: Research-problem detective

Pick a CS domain: Network security / intrusion detection

List **four possible sources** from which a research problem in this domain could emerge:

1. Literature gaps found in ML-IDS survey papers.
2. High false-positive rates reported by industry SOC teams.
3. Emerging attack types from IoT and 5G deployments.
4. Public benchmark datasets such as CICIDS2017.

C2. Identification workflow



Activity 7: Apply the four-step workflow

Topic/domain: Machine learning for network intrusion detection

1. Explore broadly: What is the larger area?

Cybersecurity and network-traffic analysis.

2. Narrow through reading: What specific sub-area or pain-point interests you?

Reducing false positives in ML-based intrusion detection.

3. Formulate question(s): Write one research question.

Can feature selection lower the false-positive rate of an ML NIDS without hurting recall?

4. Validate: What would you check for feasibility and novelty?

Dataset availability, compute cost, and how it differs from existing IDS studies.

Quick Pair Challenge: Problem or Topic?

Classify each statement as **Topic**, **Weak Problem**, or **Research Problem**.

1. "Artificial Intelligence in Smart Cities." Topic
2. "AI is interesting." Weak Problem
3. "How can reinforcement learning optimize traffic signals in smart cities?" Research Problem
4. "Cybersecurity." Topic
5. Your own statement: How can ML cut false positives in NIDS?

C3. Characteristics of a good research problem

A strong problem should be **clear and concise, researchable and feasible, relevant and original**, and should offer scope for **data collection and analysis**. The syllabus also emphasizes essential attributes and avoiding errors in selection. **[5 marks]**

Criterion	Evidence that YOUR problem satisfies it
Clear and concise	States one clear goal: cut false positives in an ML NIDS.
Researchable	Testable on public IDS datasets with standard metrics.
Feasible	Uses available datasets, ML libraries, and a single GPU.
Relevant	False alarms overwhelm analysts, so reducing them helps SOCs.
Original / non-trivial	Targets FP reduction, less studied than raw accuracy.
Allows data collection and analysis	Yields precision, recall, and FP-rate for quantitative analysis.

4 Part D — What Can Go Wrong? Scope and Boundaries

Common errors in selecting a research problem

1. Ambiguity / vagueness
The problem does not clearly state what is being investigated.
2. Too broad
The study becomes unrealistic or unmanageable.
3. Too narrow
There may be insufficient room for meaningful investigation.
4. Lack of relevance
The problem has weak academic, practical, or societal value.
5. Lack of feasibility
Resources, data, skills, time, or infrastructure are unavailable.
6. Ignoring prior art
Existing research may already answer the question.

Activity 8: Error clinic

For each weak problem, identify the main error and rewrite it.

1. "Improve AI everywhere."
   Error: Too broad / vague - unmanageable with no clear focus.
   Improved version:
   Improve CNN inference speed on edge devices for real-time image classification.

2. "Study all cybersecurity threats affecting every organization worldwide."
   Error: Too broad and infeasible in scope.
   Improved version:
   Study phishing-URL detection accuracy for a university email system over one semester.

3. "Build a system that detects every possible disease with 100% accuracy in two days."
   Error: Unrealistic and infeasible target and timeline.
   Improved version:
   Develop an ML model to grade diabetic-retinopathy severity from retinal images with high recall.

How to avoid selection errors

- Perform a literature review.
- Consult experts or guides.
- Conduct pilot studies where appropriate.
- Align the problem with objectives and available resources.

D2. Scope = the boundary of your research

Scope defines the **limits and boundaries** of the study and prevents deviation from the objectives. Typical boundaries include:

Time frame | **Geographic** location | **Target population** or dataset | **Tools /** technologies

Activity 9: Draw the boundary

Research problem: *"How can machine learning improve network intrusion detection?"*

Fill in realistic boundaries:

Time frame	3 months (one project cycle).
Geographic / deployment context	Enterprise LAN; offline dataset evaluation.
Target dataset / population	CICIDS2017 network-flow dataset.
Tools / algorithms	Python, scikit-learn; Random Forest and XGBoost.
What is explicitly OUT of scope?	Live deployment, encrypted-payload inspection, zero-day attacks.

Activity 10: Scope test

A student's project is titled *"AI for Smart Cities."* Give three reasons why this scope is likely too broad, then convert it into a narrower study.

Too broad because: (1) 'AI' covers many techniques; (2) 'smart cities' spans traffic, energy, water and safety; (3) no dataset, metric, or timeframe is fixed. Narrower study: 'Using reinforcement learning to reduce average vehicle waiting time at signalized intersections in a simulated city grid.'

5 Part E — Research Objectives and SMART Thinking

E1. Why objectives matter

Research objectives:

- set the direction and purpose of research;
- help define scope and methodology; and
- provide focus and coherence.

A good objective is not just an activity such as "study AI." It states **what will be achieved or evaluated**.



Activity 11: SMART objective laboratory

Research topic/problem: ML-based network intrusion detection with fewer false positives

Write one objective, then audit it.

Specific	What exactly will be done? Build and tune an ML classifier that flags malicious network flows.
Measurable	What variable, metric, or evidence will show achievement? Reduce the false-positive rate below 2% while keeping recall at or above 95%.
Achievable	Why is it realistic with available resources? Public datasets and standard ML libraries make it feasible on one GPU.
Relevant	How does it address the research problem? Directly tackles alert fatigue, the core problem for SOC analysts.
Time-bound	By when / over what period? Completed within a 12-week semester project.

Final SMART objective:

Within 12 weeks, develop an ML NIDS on CICIDS2017 that lowers the false-positive rate below 2% while maintaining recall of at least 95%.

Peer Review — 60-second test

Exchange your objective with a partner. Your partner must answer:

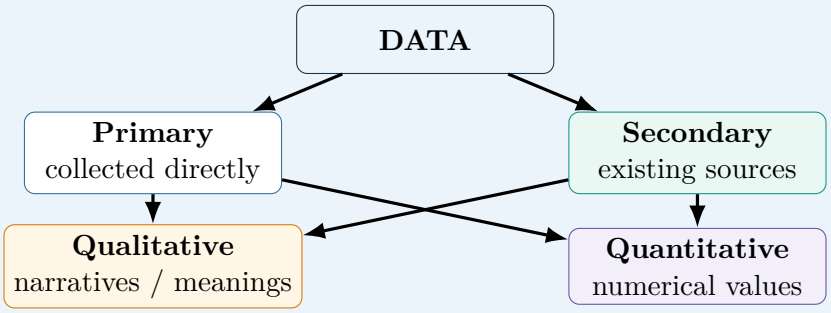
1. Can I identify exactly what is being investigated?
2. Can I identify how success will be measured?
3. Is the target realistic?
4. Does it directly connect to the problem?
5. Is a time boundary visible?

Partner's one improvement suggestion:

Specify the exact dataset and baseline model used for comparison.

6 Part F — Data Fundamentals

F1. Two useful ways to classify data



Examples discussed in class include primary interviews, sensor readings, and usage logs; secondary GitHub/Kaggle datasets, IEEE articles, and government data.

Activity 12: Classify the data

Mark each as Primary/Secondary and Qualitative/Quantitative where applicable.

Data source	Primary/Secondary	Qualitative/Quantitative
Interview transcripts	Primary	Qualitative
Sensor temperature readings	Primary	Quantitative
Kaggle dataset downloaded for analysis	Secondary	Quantitative
System CPU utilization logs collected during your experiment	Primary	Quantitative
User comments about usability	Primary	Qualitative
IEEE article used as an existing source	Secondary	Qualitative

F2. Choosing data

For your own research problem:

1. What data do you need?

   Labeled network-flow records (benign vs. attack).

2. Why is this data appropriate?

   They directly support training and evaluating an intrusion classifier.

3. Primary, secondary, or a combination?

   Combination: public datasets plus self-captured lab traffic.

4. Qualitative, quantitative, or a combination?

   Mainly quantitative (numeric flow features and metrics).

7 Part G — Data Collection Tools and Guidelines

G1. Match the tool to the evidence

The class material highlighted:

- **Questionnaires / online forms** — structured responses from participants.
- **Observation logs / experiments** — record observed behavior or experimental events.
- **System monitoring tools** — capture system/network performance.
- **Web scraping** — collect information from web sources using appropriate tools.

Examples shown in the class material include Google Forms, Typeform, Wireshark, PerfMon, BeautifulSoup, and Selenium.

Activity 13: Tool selection game

You need to collect the following evidence. Select the most suitable tool and justify.

Evidence required	Tool / method	Why?
Students' satisfaction with an app	Questionnaire (Google Forms)	Collects structured self-reported feedback at scale.
Network packet behavior during an experiment	Wireshark	Captures and inspects live packet-level traffic.
CPU/memory usage of a program	PerfMon / system monitor	Logs CPU and memory use during runs.
Behavior of users interacting with a prototype	Observation log / usability test	Records real interaction behaviour directly.
Public web data needed for a study	Web scraping (BeautifulSoup)	Automates collection of public web data.

Activity 14: Design a mini data-collection instrument

Choose one: ☐questionnaire ☐observation log ☐experiment log ☑system-monitoring plan.

Research problem: ML-based network intrusion detection

What will be measured/recorded?	Per-flow features and predicted labels (attack/benign).
Who/what is the source?	Network flows from CICIDS2017 and a lab testbed.
When/how often?	Continuously during each experiment run.
What tool will be used?	Wireshark + CICFlowMeter; Python logging.
How will consistency be maintained?	Fixed capture settings and one shared feature pipeline.

8 Part H — From Data to Meaning: Analysis and Interpretation

H1. Do not confuse these stages

Collect raw evidence → **Analyze** organize/compare → **Interpret** explain meaning → **Conclude** answer problem

Analysis asks: "What patterns, differences, relationships, or measurements are in the data?"
Interpretation asks: "What do those findings mean for my research problem and objectives?"

Activity 15: Read the result, then interpret it

Suppose an experiment gives:

Model	Accuracy	Inference time (ms)
A	92%	20
B	95%	70
C	94%	25

1. Which model has the highest accuracy?

   Model B (95%).
2. Which model is fastest?

   Model A (20 ms).
3. If the system is real-time and accuracy must be at least 93%, which model would you investigate first? Why?

   Model C: it meets the 93% bar (94%) with near-fastest latency (25 ms) - best speed/accuracy balance.
4. Is "B is best" a valid conclusion without considering the research objective? Explain.

   No - B's 70 ms latency fails real-time needs; 'best' depends on the objective, not accuracy alone.

Activity 16: Interpretation vs. description

Complete each sentence with either **description** or **interpretation**.

1. "Model B achieved 95% accuracy." description
2. "The higher accuracy may be useful when accuracy is the primary objective, but its latency may limit real-time deployment." interpretation
3. "The system processed 10,000 requests." description
4. "The observed throughput suggests the system can support the target workload." interpretation

Researcher's caution

A graph can show a pattern, but a pattern is not automatically an explanation. Write one question you would ask before claiming *why* a result occurred.

Could a confounding factor (e.g., dataset imbalance or hardware) explain the result rather than the model itself?

9 Part I — Research Instrumentation

I1. Instrumentation in computing research

Research instrumentation refers to the **tools, hardware, and software used for data collection and analysis**. Class examples include:

- sensors;
- simulators;
- benchmarking suites;
- custom tools for robotics, networks, and AI.

Two important quality ideas emphasized in the class material are **calibration** and **repeatability**.

Activity 17: Instrument selection matrix

For your chosen research problem, identify one instrument at each level.

Level	Possible instrument	What evidence will it produce?
Hardware / sensor	Network TAP / NIC in promiscuous mode	Raw packet captures from the link.
Software / simulator	NS-3 / Mininet traffic simulator	Reproducible synthetic attack traffic.
Benchmark / measurement tool	scikit-learn metrics + CICFlowMeter	Precision, recall, FP-rate, flow features.
Custom tool (if needed)	Python feature-extraction & logging script	Labeled feature vectors for training.

Activity 18: Calibration and repeatability

Why should a sensor, measurement setup, benchmark, or custom instrument be calibrated?

So its measurements are accurate and unbiased, giving trustworthy, comparable data.

What could happen if the same experiment cannot be repeated consistently?

Results become unreliable and unverifiable, so conclusions cannot be trusted or reproduced.

10 Part J — The Research Problem Builder

Your 10-minute research design

Choose a real Computer Science problem you care about. Complete every box.

Step 1 — Domain

Network security (intrusion detection).

Step 2 — Real-world pain-point

IDS raise too many false alarms, causing analyst alert fatigue.

Step 3 — Research problem statement

ML-based NIDS often trade fewer false positives for lower detection.

Step 4 — Research question

Can feature selection cut false positives without reducing recall?

Step 5 — Why is it worth investigating?

Fewer false alarms let analysts focus on genuine threats.

Step 6 — Research approach

☐Exploratory ☐Descriptive ☐Analytical ☐Theoretical ☑Empirical ☐Fundamental ☑Applied

Step 7 — Scope boundaries

Time: 3 months Dataset/population: CICIDS2017

Tools/technology: Python, scikit-learn Out of scope: Live deployment, zero-days

Step 8 — Three SMART objectives

O1: Build a baseline Random-Forest classifier on CICIDS2017.

O2: Apply feature selection to cut the false-positive rate below 2%.

O3: Compare recall and FP-rate against the baseline within 12 weeks.

Step 9 — Data plan

Data type(s): Quantitative flow features Source(s): CICIDS2017 + lab capture

Collection tool(s): CICFlowMeter, Wireshark Instrumentation: Python ML pipeline

Step 10 — Analysis and interpretation

What will you measure/analyze? How will the findings answer the research question?

Measure precision, recall, and FP-rate; a lower FP-rate at fixed recall shows the method answers the question.

11 Part K — Case Study: AI in Smart Cities

Case Study — Refine the vague idea

A student wants to research:

"Artificial Intelligence in Smart Cities."

Your task is to transform this broad idea into a defensible Unit I research design.

K1. Narrow the problem

Broad area: **AI + Smart Cities**

Choose one specific application: ☑traffic ☐waste ☐energy ☐water ☐safety ☐other: _____

Specific pain-point:

Fixed-timer signals cause congestion at variable-demand intersections.

Refined research problem:

Can reinforcement learning reduce average vehicle waiting time at signalized intersections versus fixed-time control?

K2. Define the approach

Dominant approach: Empirical (simulation-based)

Why is it theoretical, empirical, exploratory, descriptive, analytical, or a combination?

Empirical: it is tested by running RL on a traffic simulator and measuring outcomes.

K3. Three SMART objectives

1. Model a 4-intersection grid in SUMO with realistic demand.
2. Train a DQN signal controller and measure waiting time.
3. Cut mean waiting time by at least 15% vs. fixed-time within 10 weeks.

K4. Boundaries

Timeframe	10 weeks.
Target dataset / population	Simulated traffic flows (SUMO scenarios).
Geographic/deployment context	One simulated urban grid (offline).
Tools/technologies	SUMO, Python, TensorFlow (DQN).
Out of scope	Real-world deployment, pedestrians, multi-city networks.

12 Part L — Case-Based Thinking

Case 1: "Build a better chatbot"

A student proposes: *"I will make a chatbot that is better than ChatGPT."*
Identify at least **four** problems with this research proposal at the Unit I level.

1. 'Better than ChatGPT' is vague and not measurable.
2. Scope is far too broad and infeasible for a student project.
3. No specific task, dataset, or evaluation metric is defined.
4. Ignores prior art and available resources.

Rewrite it as a focused research problem:

Can a fine-tuned small LLM improve FAQ-answering accuracy for a university helpdesk versus a retrieval baseline?

Case 2: Sensor-based smart agriculture

A team wants to predict irrigation needs using soil-moisture sensors.
Identify:

1. One possible research problem.
   Can soil-moisture data predict a crop's daily irrigation need?
2. Two data sources.
   Soil-moisture sensors and local weather records.
3. One primary data source.
   Field soil-moisture sensor readings.
4. One secondary data source.
   Meteorological-department rainfall data.
5. One data-collection tool.
   IoT sensor with a data logger.
6. One research instrument.
   Calibrated soil-moisture sensor.
7. One measurable objective.
   Predict irrigation need within +/-10% over one season.
8. One scope boundary.
   One crop type on a single farm plot.

Case 3: The "perfect accuracy" trap

A researcher claims: *"My model is successful because it gives 99.9% accuracy."*
Write three questions you would ask before accepting the claim:

1. On what dataset and train/test split was 99.9% measured?
2. Is the data balanced, or does accuracy hide poor recall?
3. Does it generalize to unseen, real-world data?

13 Part M — Unit I Rapid-Fire Quiz

Choose the best answer.

1. A research problem should primarily be:

   (a) fashionable
   (c) extremely broad
   (b) clear and researchable
   (d) impossible to test

2. Which is a source of research problems?

   (a) literature gaps
   (c) avoiding prior work
   (b) random guessing only
   (d) unrelated facts

3. "Measure the average latency of Algorithm X on Dataset Y within four weeks" is closest to:

   (a) vague topic
   (c) literature review
   (b) SMART objective
   (d) research instrument

4. Existing data from Kaggle is generally:

   (a) primary
   (c) always qualitative
   (b) secondary
   (d) always experimental

5. "User comments about usability" are primarily:

   (a) qualitative
   (c) hardware
   (b) quantitative
   (d) benchmark

6. Which is most directly related to system monitoring?

   (a) PerfMon
   (c) patent claim
   (b) title page
   (d) abstract

7. A study based mainly on observation and experiments is:

   (a) empirical
   (c) descriptive writing
   (b) purely theoretical
   (d) scope definition

8. Which is NOT a good scope boundary?

   (a) dataset
   (c) tools
   (b) timeframe
   (d) "everything"

9. Interpretation primarily asks:

   (a) What does the result mean?
   (c) Who printed the report?
   (b) What font was used?
   (d) What is the page number?

10. Calibration and repeatability are especially important for:

(a) research instrumentation

(b) research titles

(c) roll numbers

(d) acknowledgements

Rapid-fire: One-word / one-line

1. Define research in your own words:

   Systematic, evidence-based investigation to answer questions or verify knowledge.

2. One difference between a topic and a research problem:

   A topic is a broad area; a research problem is a specific, answerable question.

3. One source of a research problem:

   Literature gaps.

4. One common selection error:

   Making the scope too broad.

5. Expand SMART:

   Specific, Measurable, Achievable, Relevant, Time-bound.

6. One primary-data example in CS:

   CPU-utilization logs from your own experiment.

7. One secondary-data example in CS:

   A Kaggle or IEEE dataset.

8. One data collection tool:

   Wireshark.

9. One research instrument:

   A benchmarking suite.

10. Difference between analysis and interpretation:

    Analysis finds patterns; interpretation explains what they mean.

14 Part N — Connect the Dots

Match Column A with Column B.

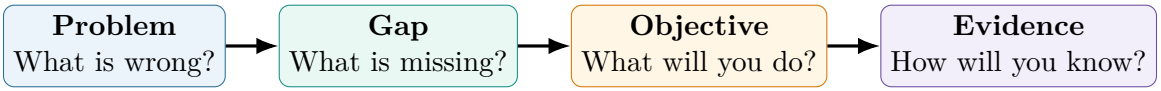
A	Concept	B	Meaning / Example
1	Exploratory	A	Collected directly
2	Primary data	B	Clear, measurable, realistic, relevant, time-bound
3	Scope	C	Discover patterns / new ideas
4	SMART	D	Limits and boundaries of a study
5	Empirical	E	Observation / experiment based
6	Instrumentation	F	Tools/hardware/software for collection and analysis
7	Secondary data	G	Existing datasets/articles/logs
8	Interpretation	H	Explain what findings mean for the problem

Answers: 1-C, 2-A, 3-D, 4-B, 5-E, 6-F, 7-G, 8-H.

15 Part O — Viva / Oral Defense Practice

Imagine your teacher asks: *"Defend your research problem in 60 seconds."*

Use the structure:



Your 60-second defense notes:
Problem: Gap: Objective: Evidence/data:

Problem: SOC analysts are flooded by false IDS alerts. Gap: Little work targets false-positive reduction. Objective: Cut the FP-rate below 2% while keeping recall at or above 95%. Evidence: precision, recall, and FP-rate on CICIDS2017.

16 Part P — Unit I Master Check

Can I do it without looking at the PPT?

Rate yourself honestly: **1 = not yet, 2 = partly, 3 = confidently**.

Skill	1	2	3
Explain what research is and why it matters			✓
Identify the key elements of research			✓
Generate sources of research problems			✓
Distinguish a topic from a research problem			✓
Evaluate a research problem using quality criteria			✓
Identify common selection errors			✓
Define scope and boundaries			✓
Write SMART objectives			✓
Choose an appropriate research approach			✓
Distinguish theoretical and empirical research			✓
Classify primary/secondary data			✓
Classify qualitative/quantitative data			✓
Select a suitable collection tool			✓
Explain analysis vs interpretation			✓
Select suitable research instrumentation			✓
Explain why calibration and repeatability matter			✓

My one-minute takeaway

The most important idea I learned in Unit I is:

A good research problem must be specific, feasible, and measurable - not just a trendy topic.

The concept I still need to revise is:

Writing fully SMART objectives quickly.

One research problem I can now formulate is:

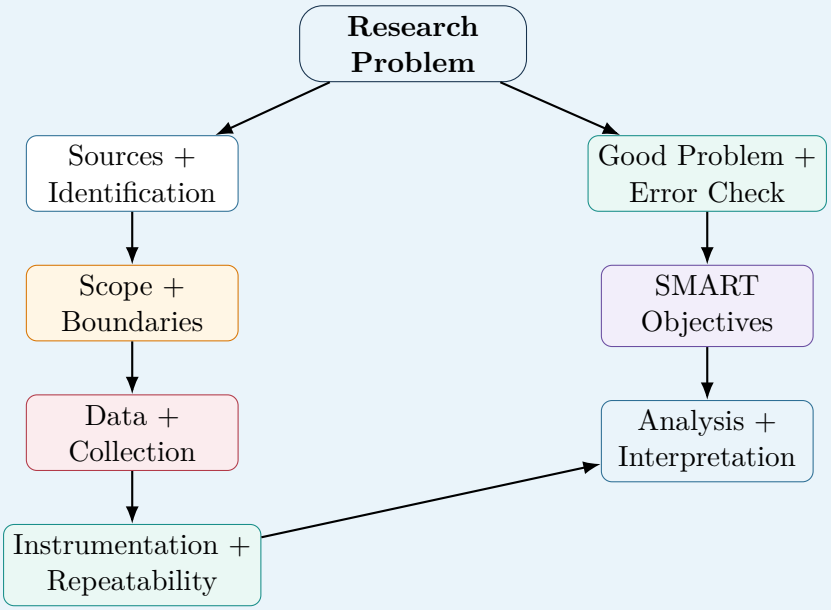
Reducing false positives in ML-based network intrusion detection.

Exit Ticket: Researcher Mindset

Complete the sentence:
"A good researcher does not start by asking which tool to build *; a good researcher first asks* what problem to solve *."*
Explain your answer in one or two lines:

Tools are only a means; research must be driven by a clear, worthwhile problem and question first.

Unit I in one visual



End of Unit I Worksheet — Introduction to Research Problem